
ENGINEERING CHEMISTRY

7-Unit Syllabus Mapping & PYQ Analysis Report

Course Code: BT-101 (GS / CBGS)

Target University: Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV)

Syllabus Alignment: 7-Unit Structure (Revised)

Analysis Coverage: 2020 – 2025 (10 Active Exam Papers)

Compiled Year: 2026

Strictly categorized according to official department modules for structured study and high-yield revision.

Table of Contents

Section 1: 7-Unit Categorized Question Bank	2
Unit 1: Water – Analysis, Treatments and Industrial Applications	2
Unit 2: Boiler problem & softening methods	3
Unit 3: Lubricants and Lubrication	4
Unit 4: Polymer & polymerization	5
Unit 5: Phase equilibrium and Corrosion	6
Unit 6: Spectroscopic techniques and application	7
Unit 7: Periodic properties	8
Section 2: High-Yield Topic Matrix (Unit-wise Breakdown)	9
Section 3: Ranked Frequency Analysis of Most Repeated PYQs	12

Section 1: 7-Unit Categorized Question Bank

Every single question from the provided 10 historical question papers has been extracted exactly as originally worded and systematically allocated into the prescribed 7-unit structure.

UNIT 1: Water – Analysis, Treatments and Industrial Applications

- Define Hardness of water. Explain the various units of hardness and their inter-relationship. Why is EDTA method preferred over other methods for hardness determination? **(Dec 2025, Dec 2020)**
- Explain the principle of EDTA titration. Calculate the total hardness (in ppm) of a water sample containing: $\text{Ca}(\text{HCO}_3)_2 = 16.2 \text{ mg/L}$, $\text{Mg}(\text{HCO}_3)_2 = 7.3 \text{ mg/L}$, $\text{MgCl}_2 = 9.5 \text{ mg/L}$ and $\text{CaSO}_4 = 13.6 \text{ mg/L}$. **(Dec 2025)**
- 100 ml water sample required 20 ml of $N/50 \text{ H}_2\text{SO}_4$ for neutralisation to Phenolphthalein end point. After Methyl orange was added to this and further acid required was 2.5 ml. Calculate the alkalinity type and extent in terms of CaCO_3 equivalent. **(June 2025, Dec 2020)**
- Calculate temporary, permanent and total hardness of water sample containing $\text{Mg}(\text{HCO}_3)_2 = 7.3 \text{ mg/L}$, $\text{Ca}(\text{HCO}_3)_2 = 16.4 \text{ mg/L}$, $\text{MgCl}_2 = 9.5 \text{ mg/L}$ and $\text{CaSO}_4 = 13.6 \text{ mg/L}$. **(June 2025)**
- What is Temporary and Permanent hardness water, How it is differentiate? Give suitable examples. **(Dec 2024)**
- Differentiate between hard water and soft water. What is meant by hardness of water? Write various units to express hardness of water giving their relationship. **(June 2024)**
- 100 ml of a water sample required 18 ml of standard EDTA solution. 20 ml of standard CaCO_3 (1.5g/L) solution required 25 ml of EDTA solution. 100 ml of the sample after boiling required 12 ml of EDTA solution. Calculate the temporary and permanent hardness of the sample. **(Dec 2023)**
- Discuss method for determination of alkalinity in given water sample. **(June 2022)**
- What is the principle of EDTA method? Explain the estimation of total hardness of water by complexometric method. **(Nov 2022)**
- Calculate the total hardness of a water sample which shows following analysis: $\text{Ca}(\text{HCO}_3)_2 = 4.86 \text{ mg/L}$; $\text{Mg}(\text{HCO}_3)_2 = 5.84 \text{ mg/L}$; $\text{CaSO}_4 = 6.8 \text{ mg/L}$ and $\text{MgSO}_4 = 8.4 \text{ mg/L}$. **(Nov 2022)**
- Explain hardness of water. How it is expressed in various units? Discuss harmful effects of hard water in boilers. *[Note: Multi-unit assignment: introductory definitions belong to Unit 1; boiler effects belong to Unit 2]* **(June 2020, June 2020-CBGS)**
- Write a short note on Alkalinity. **(Dec 2023, June 2020-CBGS)**

UNIT 2: Boiler problem & softening methods

- Compare Zeolite and Ion-Exchange processes for water softening. Discuss why Ion-Exchange is superior for high-pressure boilers. **(Dec 2025)**
- What is Desalination? Describe the Reverse Osmosis (RO) process with a neat labeled diagram. State its industrial applications. *[Note: Assigned here as an advanced terminal softening/purification methodology]* **(Dec 2025)**

- Explain the chemical reactions involved in the Lime-Soda process. Why is a small amount of coagulant (like Alum) often added during this process? **(Dec 2025)**
- What is lime-soda process? Calculate the requirement of Lime (80% pure) and Soda (90% pure) for treatment of 1,00,000 liters of water containing: $\text{Ca}(\text{HCO}_3)_2 = 40.5 \text{ ppm}$, $\text{Mg}(\text{HCO}_3)_2 = 36.5 \text{ ppm}$, $\text{MgSO}_4 = 30 \text{ ppm}$ and $\text{CaCl}_2 = 22.2 \text{ ppm}$. **(Dec 2025)**
- Write short note on: (i) Sludge and Scale, (ii) Caustic Embrittlement, (iii) Boiler Corrosion. **(June 2025, Dec 2023, June 2020)**
- Describe ion exchange process used for softening of water with neat diagram. A zeolite softener was 90% exhausted by removing the hardness completely when 10,000L of hard water sample passed through it. The exhausted zeolite bed required 200L of 3% NaCl solution for its complete regeneration. Calculate the hardness of water sample. **(June 2025)**
- Calculate the amount of lime and soda required for softening of 20,000 liters of water whose analysis is: $\text{Ca}(\text{HCO}_3)_2 = 40.0 \text{ ppm}$; $\text{Mg}(\text{HCO}_3)_2 = 36.5 \text{ ppm}$; $\text{CaSO}_4 = 34.0 \text{ ppm}$; $\text{MgSO}_4 = 30.0 \text{ ppm}$; $\text{CaCl}_2 = 27.75 \text{ ppm}$. **(June 2025)**
- Write a note on scales and sludges. **(Dec 2024, June 2024, Dec 2023, June 2020-CBGS)**
- Explain Ion-exchange process of water softening with neat diagram. **(Dec 2024, June 2023, June 2022, Nov 2022, June 2020-CBGS)**
- What are boiler troubles? How are they caused? Suggest steps for minimizing the boiler troubles. **(Dec 2024)**
- Discuss the following terms: (i) Blow down process, (ii) Carry over. **(June 2024)**
- What are sludge and scale in boilers? How are they formed? Suggest any two methods to prevent their formation. **(June 2024)**
- Discuss the zeolite process for softening of water. **(Dec 2023, Nov 2022)**
- How is water softened by the lime-soda process? What are its advantages? **(Dec 2023, June 2022)**
- Explain the formation and composition of scales in boilers. How are they different from sludge's? What are bad affects? **(June 2023)**
- A water sample contains the following impurities per liter: $\text{CaCl}_2 = 222 \text{ mg}$; $\text{Mg}(\text{HCO}_3)_2 = 73 \text{ mg}$; $\text{CaSO}_4 = 6.8 \text{ mg}$; $\text{NaCl} = 58.5 \text{ mg}$; $\text{Ca}(\text{NO}_3)_2 = 162 \text{ mg}$ and $\text{MgSO}_4 = 73 \text{ mg}$. The sample of lime available is 74% pure and soda is 90% pure. 20% excess chemicals have to be added. Calculate the amount lime and soda required for the treatment of 10000 liters of water. **(June 2023)**
- Explain the hot lime-soda process. What are its advantages? **(June 2023)**
- What is Caustic embrittlement? How do you prevent it? **(Nov 2022, June 2023)**
- How is portable water disinfected by ozonation? *[Note: Assigned here under water treatment methods]* **(Nov 2022)**
- Discuss harmful effects of hard water in boilers. **(June 2020-CBGS)**

UNIT 3: Lubricants and Lubrication

- Define Lubrication. Explain the mechanism of Extreme Pressure Lubrication. Which type of lubricants are used for heavy-duty machinery? **(Dec 2025)**
- Discuss the importance of Viscosity Index. How does it affect the performance of an engine oil in varying temperatures? **(Dec 2025, June 2024, Dec 2023)**

- An oil sample under test has a Saybolt Universal Viscosity same as that of standard Gulf oil (low viscosity standard) and Pennsylvanian oil (high viscosity standard) at **210°F**. Their Saybolt Universal Viscosities at **100°F** are 61, 758, and 420 seconds, respectively. Calculate viscosity index of the sample. **(June 2025)**
- Explain the following properties (any two): (i) Aniline Point, (ii) Cloud and Pour Point, (iii) Flash and Fire Point. **(June 2025, June 2024, Dec 2024, Dec 2020)**
- What are the important properties that an efficient lubricant should possess? 1.5 gm of a lubricant oil dissolved in alcohol require 24 ml of 0.0125 N alcoholic KOH for neutralization. What is neutralization value of the lubricant? If an oil to be used in machinery should not have neutralization value greater than 8.0, is it safe to use the test sample as a lubricant in the machinery? **(June 2025)**
- Give short note on mechanism thick film of lubrication process. **(Dec 2024)**
- What is Lubrication? Give classification of lubricants. **(June 2024, Nov 2022)**
- What is Lubricant? With the help of neat diagrams, explain the following criteria of lubricants: (i) Viscosity index, (ii) Flash and fire point. **(Dec 2023)**
- Describe the mechanism of lubrication that is applied to delicate instruments. **(June 2023)**
- Explain about aniline point and saponification Number. **(June 2023)**
- Write short notes on: Thin film lubricants and fluid film lubricants. **(June 2022)**
- Write a note on lubricants with special reference to their: (i) classification, (ii) mode of action with examples and uses. **(Nov 2022)**
- Explain following properties of lubricants giving their significance: (i) Viscosity Index, (ii) Flash point, (iii) Steam Emulsion Number. **(Dec 2020)**
- What is lubricant? Discuss following properties of lubricant giving their significance: (i) Viscosity index, (ii) Flash point. **(June 2020, June 2020-CBGS)**
- Write short note on: (i) Fire point, (ii) Emulsification number. **(June 2020-CBGS, Dec 2024)**

UNIT 4: Polymer & polymerization

- Discuss the mechanism of Addition and Condensation polymerization with one example each. Define "Functionality" of a monomer. **(Dec 2025, June 2023)**
- Write the synthesis, properties and engineering uses of: (i) Bakelite, (ii) PMMA, (iii) Neoprene. **(Dec 2025)**
- Distinguish between thermoplastic and thermosetting resins. Give preparation and application of Urea-Formaldehyde and PVC. **(June 2025, Nov 2022, Dec 2023)**
- Differentiate chain growth and step growth polymerization. Explain Free radical and Ionic polymerization with the help of suitable example. **(June 2025, June 2023, Nov 2022)**
- What are polymers? Give their classification with examples. **(Dec 2024, June 2020-CBGS)**
- Give preparation, properties and uses of Bakelite. **(Dec 2024, Dec 2023, Nov 2022)**
- Illustrate free radical mechanism of polymer. **(June 2024)**
- Explain about Biodegradable polymers with suitable example. **(June 2024, June 2022)**
- Write a short note on synthetic rubber with a suitable example. **(Dec 2023)**
- Write a note on preparation, properties and uses of the following: (i) Phenol-formaldehyde resin, (ii) Buna-S. **(June 2023)**

- What are living polymers? Give preparation, properties and uses of PVC. **(June 2022)**
- What is Elastomers? Give the preparation and properties and uses of BUNA's. **(June 2022)**
- Write short note on: Conducting polymers. **(Dec 2024)**
- What is Polymerisation? Discuss mechanism of Polymerisation. Write Preparation, properties and uses of following: (i) PVC, (ii) Nylon 6:6, (iii) Poly Ethylene. **(Dec 2020)**
- Write short note on: Vulcanisation of Rubber. **(Dec 2020, June 2022)**
- What are polymers? Give classification of polymers. Discuss mechanism of polymerisation. **(June 2020-CBGS)**

UNIT 5: Phase equilibrium and Corrosion

- Explain the "Differential Aeration Theory" of corrosion with the help of Pitting Corrosion example. Why do iron nails rust faster in saline water? **(Dec 2025)**
- What is Sacrificial Anodic Protection? How is it used to protect underground pipelines? Compare it with Impressed Current Cathodic Protection. **(Dec 2025)**
- Define Corrosion. Write types and prevention of corrosion. Explain theory of mechanism of corrosion. How much rust ($\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$) will be formed, when 100 kg of iron have completely rusted away? (Mol. Wt. of rust 214 g). **(June 2025)**
- Define and explain the terms involved in the phase rule. Draw a neat labelled phase diagram of water system and explain the areas and curves in it. What is the significance of the triple point and meta stable curve in this system? **(June 2025, June 2023, June 2022)**
- What is oxidation corrosion and how does it take place? Describe the mechanism of oxidation corrosion. **(Dec 2024, June 2024, June 2022)**
- Explain water line corrosion and galvanic corrosion. **(Dec 2024, June 2024)**
- Write brief note on: Phase diagram of single component systems. **(June 2024, Dec 2020, Nov 2022)**
- What is meant by a phase diagram? Explain two-component system by taking Cu-Ag system as an example. **(Dec 2023)**
- Describe the following with suitable examples: (i) Chemical corrosion and Electro chemical corrosion, (ii) Phase diagram of one-component water system. **(June 2023)**
- Write short note on: Eutectic point. **(June 2023)**
- Explain pitting corrosion and stress corrosion. **(June 2022)**
- Differentiate the following with suitable examples: Galvanic series and Electrochemical series. **(Nov 2022)**
- Explain Phase Rule. Discuss phase diagram of single component system. **(Dec 2020)**
- Write short note on: Rusting of Iron. **(Dec 2020)**
- What is corrosion? Discuss various theories to explain corrosion with suitable examples. How corrosion may be prevented? **(June 2020, June 2020-CBGS)**
- Explain electrochemical theory of corrosion with reference to iron. Discuss process of galvanisation to protect iron. **(June 2020-CBGS)**
- Write short note on: Phase Equilibrium. **(June 2020-CBGS)**

UNIT 6: Spectroscopic techniques and application

- State the selection rules for IR Spectroscopy. Explain why some molecules are "IR Active" and some are not. Describe the various types of molecular vibrations. **(Dec 2025)**
- Explain the principle of Thin Layer Chromatography (TLC). How is the R_f value calculated and what does it indicate? [Note: Assigned to Unit 6 as a foundational modern experimental analytic technique] **(Dec 2025)**
- How can a combination of UV and IR spectroscopy be used to determine the structure of an unknown organic compound? Illustrate with an example. **(June 2025)**
- What is Electronic spectroscopy? An unknown substance absorbs 40% of the light of wavelength 43.5 nm in a cell of 2 cm dimension when has a concentration of 20 gL^{-1} . Under the similar conditions, what percentage of light will be absorbed by a solution containing 40 gL^{-1} of the same unknown substance? **(June 2025)**
- What is 'spectroscopy'? Discuss about Ultra-violet spectroscopy principle, instrumentation and applications. **(Dec 2024)**
- What is Spectroscopy? Describe about IR spectroscopy by giving principle, instrumentation and applications. **(June 2024, Dec 2020)**
- Write brief note on: Electronic shifts / transitions. **(June 2024, Nov 2022)**
- State and explain Beer-Lambert's Law. **(Dec 2023, June 2023, Nov 2022)**
- Define the following terms: (i) Bathochromic shift, (ii) Hypsochromic shift, (iii) Hyperchromic shift, (iv) Chromophore. **(Dec 2023)**
- Discuss principle, instrumentation and applications of vibrational and rotational spectroscopy. **(June 2023, Nov 2022, June 2020)**
- What is Electronic spectroscopy? Discuss about electronic transitions and applications of U.V. Spectroscopy. **(June 2022)**
- Write short note on: Colorimetry. **(Dec 2020)**

UNIT 7: Periodic properties

- What is periodic table? Write trends of following in a group and in a period giving suitable reasons: (i) electronegativity, (ii) electron affinity, (iii) atomic size. **(Dec 2025, June 2024, Dec 2023)**
- Write brief note on (any two): (i) Effective nuclear charge, (ii) Polarizability, (iii) Oxidation states. **(Dec 2025, June 2025, June 2023, Dec 2024)**
- Write brief note on: Aufbau rule. **(June 2025)**
- Write short notes on Electro electronegativity variation in groups and periods. **(Dec 2024)**
- Explain about polarisability. **(Dec 2024)**
- How do you explain electro negativity, ionic sizes and electron affinity of s, p - block elements? **(Dec 2023, June 2023)**
- Explain following terms for elements giving factors affecting and Periodic trends: (i) Atomic size, (ii) Electron affinity, (iii) Ionisation Energy. **(Dec 2020)**

Section 2: High-Yield Topic Matrix

This matrix maps core syllabus concepts according to exam frequency patterns and provides an engineering overview of critical high-yield checkpoints.

Units 1 & 2: Water Chemistry and Boiler Systems

Syllabus Module	Priority Class	Yield Context & Core Focus Requirements
EDTA titrations & hardness mathematical sets	★★★★ High	Calculates total, permanent, and temporary parts. Requires familiarity with Buffer systems and EBT indicator mechanisms.
Ion-Exchange & Softening Plant configurations	★★★★ High	Requires a complete schematic diagram showing separate cation and anion beds. Focuses on total mineral removal for high-pressure installations.
Lime-Soda quantitative calculations	★★★★ High	Calculates pure chemical masses considering reagent impurities and standard conversion equivalents.
Boiler failures & internal treatments	★★★ Medium	Focuses on the structural causes and prevention of scale/sludge insulation layers and caustic cracks.
Alkalinity dual-endpoint evaluations	★★★ Medium	Determines combinations of OH^- , CO_3^{2-} , and HCO_3^- using neutralizations.

Unit 3: Lubrication Engineering

Syllabus Module	Priority Class	Yield Context & Core Focus Requirements
Viscosity Index calculations & standards	★★★★ High	Evaluates viscosity-temperature profiles using standard Low (L) and High (H) reference oils.
Flash / Fire Point safety limits	★★★★ High	Essential flash prevention metric for volatile operational settings. Tested via Pensky-Marten setups.
Boundary & extreme pressure systems	★★★ Medium	Focuses on the protective chemical layers formed on machinery under extreme friction conditions.
Aniline point & critical temperatures	★ Low	Indicates aromatic content thresholds to prevent seal swelling.

Unit 4: Macromolecular Science (Polymers)

Syllabus Module	Priority Class	Yield Context & Core Focus Requirements
Bakelite / Phenol-Formaldehyde synthesis	★★★★ High	Focuses on cross-linked structural properties and thermal stability under curing.
Free-radical propagation mechanisms	★★★★ High	Covers the chain steps of peroxide initiation, chain growth, and terminal disproportionation.
PVC & structural thermoplastics	★★ Medium	Commercial synthesis lines, insulation properties, and the role of plasticizers.
Sulfur vulcanization links	★★ Medium	Explains how sulfur cross-links cross-connect elastomer chains to reduce stickiness and improve elasticity.

Unit 5: Phase Laws & Metallic Degradation

Syllabus Module	Priority Class	Yield Context & Core Focus Requirements
Water system phase diagrams	★★★★ High	Applies the Gibbs Phase Rule ($F = C - P + 2$) to maps, boundary curves, and triple equilibrium points.
Differential aeration theory (Pitting)	★★★★ High	Explains how uneven oxygen distribution creates localized anodic regions, leading to pitting defects.
Active cathodic mitigation schemes	★★ Medium	Compares sacrificial magnesium blocks with external impressed currents for pipeline protection.
Cu-Ag binary eutectic mixtures	★ Low	Analyzes two-component solid-liquid shifts and invariant eutectic points ($F' = 0$).

Units 6 & 7: Spectroscopy & Atomic Trends

Syllabus Module	Priority Class	Yield Context & Core Focus Requirements
Beer-Lambert optical attenuation profiles	★★★★ High	Derives concentration metrics based on absorption path lengths. Includes colorimetric parameters.
IR vibrational selection rules	★★★★ High	Molecules must undergo a net change in dipole moment during vibration to be IR active. Explains bending and stretching modes.
Periodic electronegativity & size variations	★★★★ High	Maps elements' properties across groups and periods using effective nuclear charge and shielding factors.

Syllabus Module	Priority Class	Yield Context & Core Focus Requirements
Electronic orbital shifts & chromophores	★★ Medium	Covers UV-induced $\pi \rightarrow \pi^*$ / $n \rightarrow \pi^*$ transitions and structural red/blue wavelength shifts.

Section 3: Ranked Frequency Analysis of Most Repeated PYQs

This section analyzes and ranks the recurring questions from the 10 exam sessions, showing exact frequency statistics and specific topic focus areas.

Rank	Consolidated Examination Question Statement	Identified Session Occurrences	Count
1	Ion-Exchange Softening System: Describe the process with a clear diagram, active chemical equations, and its role in protecting high-pressure boilers.	Dec 2025, June 2025, Dec 2024, June 2023, June 2022, Nov 2022, June 2020-CBGS	7 Times
2	Periodic Table Property Profiles: Trace group and period variations for electronegativity, electron affinity, and atomic dimensions with physical explanations.	Dec 2025, June 2025, June 2024, Dec 2023, June 2023, Dec 2020	6 Times
3	Gibbs Water Component Network: Detail the curves, regions, triple point coordinates, and degree of freedom calculations for the single-component water system.	June 2025, June 2023, June 2022, June 2024, Nov 2022, Dec 2020	6 Times
4	EDTA Complexometric Hardness Analysis: State the titration principles, chemical structures of indicators, and why this method is preferred.	Dec 2025, Nov 2022, Dec 2020, June 2020	4 Times
5	Boiler Corrosion & Internal scale deposits: Compare scale and sludge characteristics, caustic embrittlement conditions, and operational prevention steps.	June 2025, Dec 2023, June 2023, June 2020	4 Times
6	Bakelite Phenolic Resin Synthesis: Explain the preparation, cross-linking steps, mechanical features, and industrial applications of Bakelite.	Dec 2025, Dec 2024, Dec 2023, Nov 2022	4 Times
7	Alkalinity Experimental Determinations: Differentiate and calculate alkalinity using phenolphthalein and methyl orange endpoints.	June 2025, June 2022, Dec 2020, June 2020	4 Times
8	IR Spectroscopic Activation Conditions: Explain dipole selection criteria, active vibrational types, and operational layouts.	Dec 2025, June 2024, Dec 2020, June 2020	4 Times
9	Lubricant Performance Testing: Focuses on flash, fire, and aniline points alongside cloud and pour thresholds.	June 2025, June 2024, Dec 2024, June 2023	4 Times

Rank	Consolidated Examination Question Statement	Identified Session Occurrences	Count
10	Beer-Lambert Quantitative Absorption Laws: Formulate, derive, and explain the boundaries of the Beer-Lambert law for chemical validation.	Dec 2023, June 2023, Nov 2022	3 Times